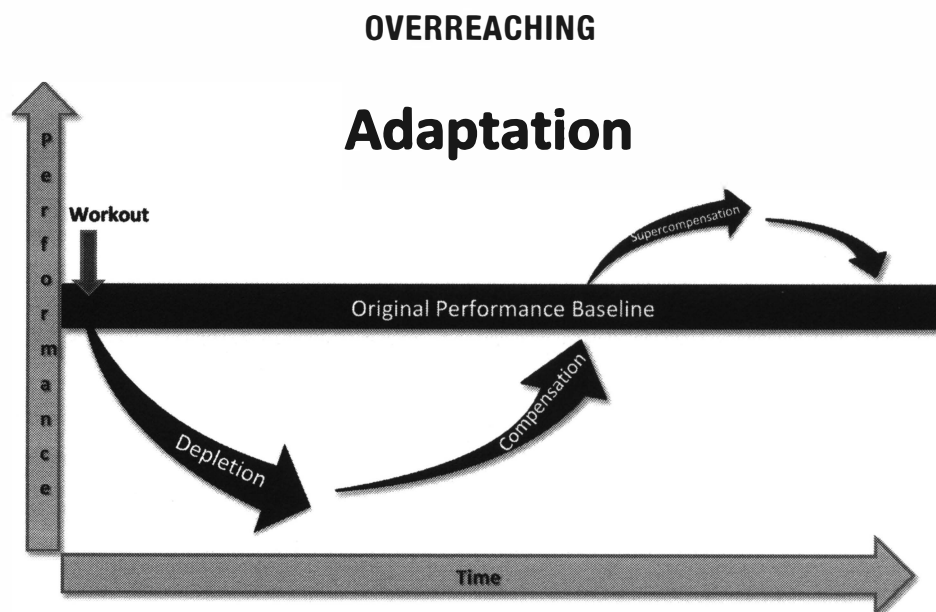


OVERREACHING AND OVERTRAINING

It is important to know the difference between *overreaching* and *overtraining*. You must learn what to watch out for and how to deal with these issues properly when they arise.

- Overreaching is a temporary state attained over multiple workouts when your abilities become depressed below baseline. A normal amount of rest or a deload period will cause your body to rebound (or supercompensate) such that your capabilities increase.
- Overtraining is a state that is reached through many workouts where your abilities have been continually depressed. This results in an overall regression of your capabilities, even after deloading and resting. Symptoms include regression of capabilities, sleep disruption, decrease in appetite, perpetual soreness, and increased risk of injury.



Overreaching is not a bad thing in itself. If you continue training for long periods of time while overreaching, however, it could devolve into overtraining. On a positive note, it is quite difficult to reach a state of

overtraining, especially as a beginner. Overreaching is much more common, and it can actually be a good thing. Many athletes plan short periods of overreaching to induce supercompensation. In sports, this is called peaking or tapering, which is usually used for competitions. Think back to the fatigue and fitness model. As fitness improves, it becomes substantially harder to progress because your body adapts to the training stimuli. Therefore, as you advance into the intermediate and advanced levels of training, you need to consider your mesocycles in terms of several weeks or two-week blocks.

The comparison to an accumulation and intensification model is obvious. An accumulation phase is essentially a carefully planned overload of your body with enough volume to depress your capabilities to a certain extent. Once this capability depression quota is hit, you shift your training to a state of intensification, where you perform a lighter volume of work at a higher intensity. Lowering the volume allows your body to rapidly supercompensate, even at a high intensity, which will reveal gains and further increase your strength.

This is the core of the “dual factor” theory which uses planned overreaching to progress. In this method, the athlete combines several workouts (typically weekly or bi-weekly) to depress the body’s capabilities and then allows progression in the following weeks. For elite athletes, planned cycles could extend over months or even over years.

Beginners can use a weekly cycle. Planned overreaching works best with at least a couple of rest or light days so that the body can recover from the total volume of the week. One of the ways you can structure a planned overreaching cycle within a week is a 4/3 (four days on; three days off), 5/2 (five days on; two days off), or similar schedule. This method is not fully recommended, as there are far easier ways to progress when you are a beginner moving into the intermediate level. (for example, simple methods of progression and light/heavy). However, this program could be useful if you are a shift worker or need your weekends free for other activities.

Thinking long-term is best when structuring planned overreaching. In an intermediate or advanced training cycle, a decrease in capabilities during the week is not always bad, provided that you have sufficient rest at the end of the week for recovery, supercompensation, and adjustment before the next few weeks of the mesocycle. An accumulation phase, where you intentionally aim to decrease your capabilities with an intensification cycle during the week after your body has recovered, may last two weeks. This would form a three-week “cycle” and you would have two of these in a six-week mesocycle.

- Week 1 – Accumulation
- Week 2 – Accumulation
- Week 3 – Intensification
- Week 4 – Accumulation
- Week 5 – Accumulation
- Week 6 – Intensification

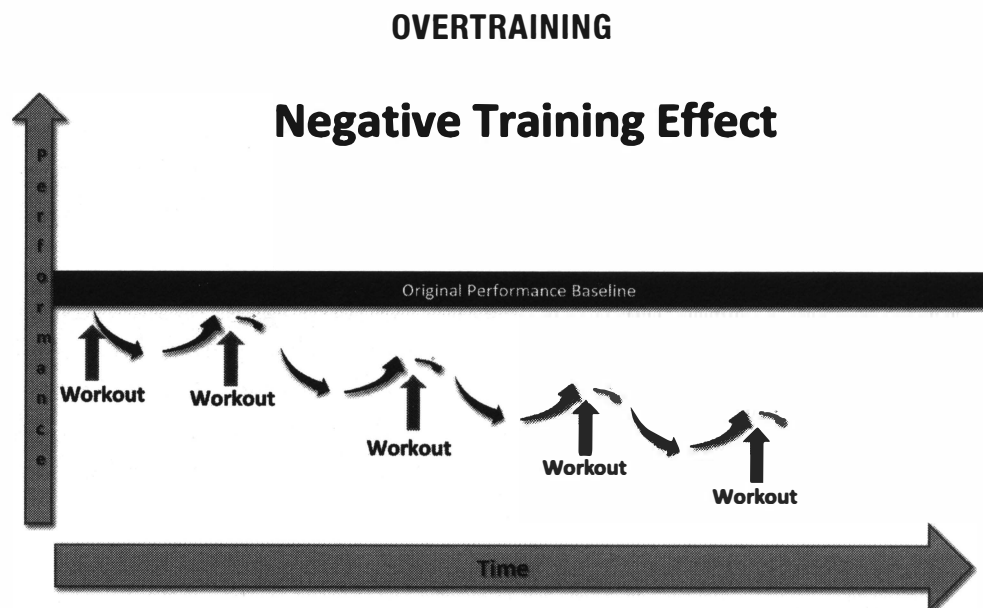
Again, it is okay if you are not progressing within a mesocycle, as gains can be masked by fatigue. Once you reach the end of the mesocycle and rest or decrease volume, your body may supercompensate and come back stronger.

Programming these advanced training concepts into your routine will definitely take thought and experimentation because everyone is different. Do not worry if you are having problems adjusting. A training log is

your best friend here, as it allows you to look back and see if your abilities were increasing or decreasing over a mesocycle. Plus, you can see the effect of supercompensation if you did not make progress within a mesocycle but suddenly got stronger after a deload week. It is important to know how your body responds at certain frequencies, intensities, and volumes of training. Becoming familiar with your own responses will help you to self-modify your routine to elicit the best progress for your level of ability.

Be aware of the differences between DUP and overreaching: you are not required to have depressed capabilities during training in order to progress. Overreaching is merely a method of applying the fatigue and fitness model to your training. DUP and other periodization models allow you to progress without depressing your abilities, even with advanced athletes. You should choose the method that best fits your overall goals.

Most high-level programs have some periodization with a type of overreaching effect built in. This allows an athlete to progress within the training sessions, as well as supercompensate to peak for competitions. However, this takes much more planning than what is covered in *Overcoming Gravity*. You have to really know how you respond to training to build in progress during a cycle *and* an overreaching effect at the end. The importance of logging your workouts cannot be overstated.



Overtraining is actually a prolonged state of “under-recovery” where your body does not have the ability to properly repair itself. It takes weeks or even months to recover from this state; your capabilities are heavily depressed, often even below starting level. This state can only be reached through chronic overworking, so you will hopefully never reach it—the only way to get to this state is by not getting enough rest for several consecutive months. Deload properly and you will have nothing to be concerned about.

The biggest thing that beginners should be concerned about is not overtraining, but performing too much volume in a single workout. This high volume is not necessarily more than a beginner can handle, but it can cause overuse injuries or inhibit optimal progression. Remember, more is not always better.

If your capabilities are increasing with each workout, you are not overreaching or overtraining. You could actually be performing either more or less than your body can handle. Experiment with volume, intensity and/or frequency of workouts to increase gains and promote recovery.

If you are plateauing or regressing and you have not taken a break from working out in a while, it may be time to deload (depending on where you are in your mesocycle). Rest to see if that helps. If not, the cause is most likely undertraining. Also be sure to examine your sleep schedule, diet, and any stress factors in your life, as all of these substantially affect recovery. Any one of these could be the actual issue. Take an extra rest day if needed to determine what is taking place. Missing a workout or two is much better than wasting time in a chronic plateau or regression.

The two most obvious symptoms of overtraining are a decrease in appetite and sleep quality. If this occurs after you have recently increased volume or raised intensity, it may be a good idea to remove any extra stress factors so your body has time to adapt. As previously stated, overtraining is not something you should be concerned about if you are progressing toward your goals. If you are not, definitely evaluate your training, sleep, diet, stress levels, and other factors. If you need advice to break through a plateau, do not hesitate to talk to someone with more experience than you.

Lyle McDonald of *Body Recomposition* has written a solid series of article on overtraining. According to observed symptoms, there are two types of overtraining—one is parasympathetic nervous system dominant (*addisonic overtraining*) and the other is sympathetic nervous system dominant (*basedowic overtraining*). The terms addisonic and basedowic are not used much anymore because the symptoms are named after specific unrelated diseases. Conceptually, however, it helps to understand them based on the symptoms of autonomic nervous system dysfunction present. Causes can be multiple—encompassing training and nutritional caloric intake as well as vitamin and mineral sufficiency, sleep, stress, and many other factors. What is important is to recognize the symptoms of each, which may give you an indication that you need to cut back on training and increase recovery:

Addisonic Overtraining

- Increased diastolic blood pressure (over 100mm/hg) during and after physical exertion.
- Impaired coordination
- Decreased endurance
- Low resting heart rate
- Persistent muscle aches
- Slight increase in recovery time
- Slight loss in motivation
- Decreased adrenal function/adrenal fatigue

Basedowic Overtraining

- Slight increase in blood pressure
- Impaired coordination with increased reaction time (reacting takes longer)
- Decreased endurance with tendency to fatigue easily
- Increased sleep requirements

- Elevated resting heart rate
- Increased appetite
- Increased metabolism and sweating
- Increased rate of breathing under stress
- Increased head aches and colds; decreased immune system
- Increased nervousness, unease, decreased motivation, eventual depression
- Increased sympathetic nervous system activity (fight or flight)

RATE OF PERCEIVED EXERTION AND TRAINING LOGS

Rate of perceived exertion (RPE) is one of the scales used to track progress in training. It is especially useful in any training that relies on auto-regulation. The RPE scale runs from 0-10 where 0 is rest and 10 is maximal exertion. You rate an exercise on an intensity range of 1-10, depending on how intense you felt the exercise was to execute on a given day. There are many types of scales that can be used for measurement (such as the *Borg*, *Visual Analog*, and *Likert*) but the RPE scale is both the simplest and the most widely used.

The RPE can tell you very quickly when you are having potential issues with an exercise. If you are performing three sets of five tuck planche pushups at an RPE of eight for the day, you will know that you can push yourself harder in the next workout. On the other hand, if your tuck planche pushups is an RPE of ten for three sets of five, you know that if you make the full jump to 3x6 tuck planche pushups in your next workout, you will likely fail. A linear repetition addition may be a better option to progress: 5-5-5 → 6-5-5 → 6-6-5 → 6-6-6.

RPE was intentionally left out of the chapter that discusses methods of strength progression because it is best used to manage and avoid overreaching and overtraining as opposed to progressing with strength workouts. For instance, you may have subsequent workouts where you perform 5-5-5 for one-arm chin-ups in a set with an RPE of ten. In your next workout, you may only hit 5-5-4 and 5-4-3 repetitions of one-arm chin-ups with the RPE holding at ten. This is a serious issue, as it indicates that you are decreasing in strength while your RPE is maxed out. You may be overreaching or overtraining. The solution in this case will likely be a deload, not increasing volume.

It is vital to use both a training log and an indicator scale like RPE. Training logs allow you to track everything related to your workout. They allow you to review your previous workouts to note when and how you progressed so you can plan solid progressions for future workouts. You may also notice patterns—both good and bad—that will help you as you construct future workout routines.

If you are a coach, it is important to maintain individual training logs for each of your athletes. This will help you see which concepts work the best for certain individuals—what works well for one athlete may impair progression with another. When coaching many athletes it is hard to remember specifics about each of their workouts from week to week; in this instance a training log is critical.

Having a training log also allows an athlete to ask for help if they are stuck. It provides a large amount of information that enables troubleshooting of a routine in all aspects: from athlete to coach, from coach to coach, from coach to athlete, and from athlete to athlete. When athletes tell their coaches that they are not progressing with their routine, the coach needs to know all of the available factors in order to make an

educated guess as to why they are not progressing. This includes the athlete's routine, sets and repetitions, tempo, rest times, overall volume, frequency, and intensity of exercises, nutrition, sleep schedule, stress levels, any potential injuries, and any other factors that may be negatively affecting their exercise program. This level of detail helps immensely in making recommendations for improvement. Attempting to do this blind is like throwing dirt against a wall and expecting it to stick, whereas if you have a detailed training log it functions like a laser-guided missile that can easily root out the areas of deficiency.

When an athlete commits to tracking their own progress, they become truly invested in their training. They will ask more questions and provide better feedback than they would if they did not keep a training log. It provides a clear pathway to mentorship in life outside the sport, which is one of the most overlooked aspects of training. Many athletes are mentored in the areas of physical development and spots, without receiving personal mentorship in life. Adding this holistic aspect to the coach-athlete relationship will help the athlete be a success in the long run of life, even outside of their sport.

CHAPTER 14 SUMMARY

OVERREACHING AND OVERTRAINING

In beginners, overreaching results in a visible stalling of progress from workout to workout. The advised remedy is to deload until you can move back to constant progression. For intermediate and advanced strength levels (where progress occurs more on a weekly basis than workout basis), planned overreaching can and should be used to apply adequate stress to the body to force strength and muscle mass adaptations.

Overtraining is not something to be concerned about unless your abilities are constantly regressing. If that is the case, it may be a good idea to take significant time off to recover—and there are serious signs and symptoms to know. Make sure you keep up with recovery factors. Sleep, nutrition, and stress reduction are the keys to get out of an overtraining deficit. If you need help, contact a medical professional.

One very good practice is to use the rate of perceived exertion (RPE) scale to track progress in a training log. Noting the RPE in addition to your repetitions, sets, volume, intensity, and frequency may be beneficial in the long run, even if it is subjective.